Problems 13

بسم الله الرحمن الرحيم

1. When Solution A, 0.1 M NaCl and Solution B, 0.20 M NaCl are separated by a semipermeable membrane, what occurs during osmosis?

A. solvent molecules move from B into A.

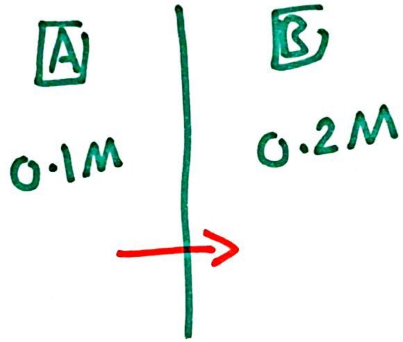
B. Na^+ and Cl^- ions move from B into A.

C. the vapor pressure of A increases.

D. the molarity of A increases.

E. no change is observed.

$$\uparrow M = \frac{n \text{ solute}}{V \downarrow}$$



osmosis

solvent → higher solvent → lower solv

Lower solu → higher solu

2. What is the mole fraction of water in a water–ethanol solution that is 54% water by mass? (Ethanol is C_2H_5OH, molar mass = 46 g/mol)

A. 0.25 B. 0.33 C. 0.46 D. 0.67 $\boxed{\text{E. } 0.75}$

100

54g H_2O → $n = \frac{54}{18} = 3$ mole

46g ethanol → $n = \frac{46}{46} = 1$ mole

$$X_{H_2O} = \frac{3}{4} = 0.75$$

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3. What is the mass percent of Na_2CO_3 (106 g/mol) in a $\boxed{1.50\ m}$ aqueous Na_2CO_3 solution?

A. 9.6 B. 10.6 C. 11.6 D. 13.7 E. 12.7

$$m = \frac{n\ Na_2CO_3}{kg\ H_2O} \leftarrow 1kg$$

$n\ Na_2CO_3 = 1.5$ mole

$$\text{Mass} = n \times Mm$$
$$= 1.5 \times 106$$
$$= 159g$$

$$\text{mass percent } Na_2CO_3 = \frac{159}{1000+159} \times 100\%$$
$$= 13.7\ \%$$

4. If 12.8 g naphthalene, $C_{10}H_8$, (M. Mass = 128 g/mol) is dissolved in 200.0 g chloroform, what is the molality of the solution?

A. 0.100 B. 0.500 C. 1.00 D. 2.00 E. Cannot determine

$n = \frac{12.8}{128}$

$= 0.1 \text{ mol}$

$m = \frac{n}{\text{kg solv}}$

$= \frac{0.1}{200 * 10^{-3}} = 0.5 m$

5. Which of the following solutes dissolved in 1000 g of water would provide a solution with the LOWEST freezing point?

A. 0.030 mole methanol, CH_3OH
B. 0.030 mole acetic acid, CH_3COOH
C. 0.030 mole ammonium nitrate, NH_4NO_3
D. 0.030 mole calcium sulfate, $CaSO_4$
E. 0.030 mole barium chloride, $BaCl_2$

CH_3OH	1	0.03
CH_3COOH	1	0.03
NH_4NO_3	2	0.06
$CaSO_4$	2	0.06
$BaCl_2$	3	0.09

$\pi V = nRT$

6. A physiologic saline solution is 0.90% by weight NaCl in water, which corresponds to a concentration of 0.154 M NaCl. At body temperature of 37 °C, what is the osmotic pressure of a physiologic saline solution? (R = 0.0821 L. atm/mol. K)

A. 16 atm B. 1.4 atm C. 0.15 atm D. 7.8 atm $\boxed{\text{E. 3.9 atm}}$

100 → 0.9 NaCl, 99.1 H_2O

$$n = \frac{0.9}{58.44} = 0.0154 \text{ mol}$$

$$M = \frac{n}{V}$$

$$0.154 = \frac{0.0154}{V}$$

$$V = 0.1 \text{ L}$$

$$\pi = \frac{nRT}{V}$$

$$= \frac{0.0154 \times 0.0821 \times (37 + 273.15)}{0.1}$$

$$= 3.9 \text{ atm}$$

7. What is the molar mass of an aromatic hydrocarbon if 0.85 g of it depresses the freezing point of 100.0 g of benzene by 0.47 °C? (K_f = 5.12 °C/m.)

A. 78 g/mol B. 93 g/mol C. 110 g/mol D. 120 g/mol E. 140 g/mol

$\Delta T_F = K_F \times m$

$0.47 = 5.12 \times m$

$m = 0.0918$

$m = \frac{n_{solute}}{\text{kg solvent}}$

$0.0918 = \frac{n}{0.1}$

$n = 9.18 \times 10^{-3}$

$n = \frac{\text{mass}}{M.m}$

$Mm = \frac{0.85}{9.18 \times 10^{-3}}$

$= 92.5957$

$\approx 93 \text{ g/mol}$

8. The molarity of a solution is defined as the

A. moles of solute per kilogram of solvent.

$\boxed{\text{B.}}$ moles of solute per liter of solution.

C. moles of solute per liter of solvent.

D. grams of solute per liter of solution.

E. grams of solute per kilogram of solvent.

$$M = \frac{n_{\text{solute}}}{V(L)_{\text{solution}}}$$

9. If 0.100 mol of chloroform, $CHCl_3$, is dissolved in 400.0 g of toluene, $C_6H_5CH_3$, the molality is

A. 0.250 B. 0.500 C. 0.750 D. 1.00 E. 2.00

$$m = \frac{n\ \text{solute}}{\text{kg solvent}} = \frac{0.1}{0.4} = 0.25$$

10. What is the molality of ethyl alcohol, C_2H_5OH, in an aqueous solution that is 50 % ethyl alcohol by mass?

A. 11 B. 15 C. 18 $\boxed{\text{D. 22}}$ E. 33

100g

50g E.A

50g H_2O

$n = \frac{50}{46.07}$

$= 1.0853$ mole

$m = \frac{1.0853}{50 \times 10^{-3}}$

$= 21.706$

≈ 22

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11. Determine the freezing point of a 0.25 m solution of glucose in water. (K_f for H_2O is 1.86 °C/m)

A. 0.93 °C   B. –0.93 °C   C. 0.46 °C   D. –0.46 °C   E. 0.23 °C

$\Delta T_F = k_f \cdot m$

$= 1.86 \cdot 0.25$

$= 0.465\ C°$

$\Delta T_F = T_{F_{solv}} - T_{F\,soln}$

$0.465 = 0 - T_{F\,soln}$

$\boxed{T_{F_{soln}} = -0.465}$

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$\Pi V = nRT$

12. Which of the following solutions would have the highest osmotic pressure?

A. 0.15 M NaCl, sodium chloride

$\boxed{\text{B}}$ 0.15 M $CaCl_2$, calcium chloride

C. 0.20 M CH_3COOH, acetic acid

D. 0.20 M $C_6H_{12}O_6$, glucose

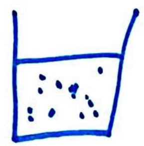
E. 0.20 M $C_{12}H_{22}O_{11}$, sucrose

NaCl	2	0.3
$CaCl_2$	3	0.45
CH_3COOH	1	0.2
$C_6H_{12}O_6$	1	0.2
$C_{12}H_{22}O_{11}$	1	0.2

13. As the number of solute particles in a given volume of solution increases,

A. the freezing point will increase and the vapor pressure will increase.
B. the freezing point will increase and the vapor pressure will decrease.
C. the boiling point will increase and the vapor pressure will increase.
D. the boiling point will decrease and the vapor pressure will decrease.
E. the boiling point will increase and the vapor pressure will decrease.

E



14. The mole fraction of NH_4Cl in a solution is 0.0311. What is the molality?

A. 1.78 m B. 1.66 m C. 0.969 m D. 0.0983 m E. 0.562 m

$$X = \frac{n_{NH_4Cl}}{(n_{NH_4Cl} + n_{H_2O})}$$

$$0.0311 = \frac{0.0311}{0.0311 + 0.9689}$$

$$\text{molality} = \frac{0.0311}{(17.4402 \times 10^{-3})}$$

$$= 1.783$$

$$n_{H_2O} = \frac{m}{M.m}$$

$$0.9689 = \frac{m}{18} \Rightarrow m = 17.4402\,g.$$

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15. If the mole fraction NaCl in a solution is 0.0175, what is the mass percent NaCl?

A. 5.77% B. 5.46 % C. 10.2% D. 11.5% E. 17.7%

$$X_{NaCl} = \frac{n_{NaCl}}{n_{NaCl} + n_{H_2O}}$$

$$0.0175 = \frac{0.0175}{0.0175 + 0.9825}$$

$$NaCl\,\% = \frac{1.0227}{17.685 + 1.0227} * 100\%$$

$$= 5.4667\,\%$$

mass NaCl $= 0.0175 * 58.44$

$= 1.0227g$

mass $H_2O = 0.9825 * 18$

$= 17.685\ g$

16. What is the mole fraction K_2SO_4 in a solution which is 3.4 m?

A. 0.0537 B. 0.0552 C. 0.0564 D.0.0584 E. 0.0598

$$\text{molality} = \frac{n_{solute}}{\text{kg solvent}} \leftarrow 1\text{ kg}$$

$$n_{K_2SO_4} = 3.4 \text{ mole}$$

$$n_{H_2O} = \frac{1000}{18} = 55.56 \text{ mole}$$

$$X = \frac{3.4}{3.4 + 55.56} = 0.05767$$

17. A 1.34 M $NiCl_2$ solution has a density of 1.12 g/cm³ (mL). What is the molality of the solution?

A. 0.913 m B. 1.55 m C. 1.42 m D. 2.55 m E. 3.13 m

$M = \frac{n_{solute}}{V_{solu}}$ ← 1L

$n_{NiCl_2} = 1.34$ mole

$d = \frac{m}{V}$

$1.12 = \frac{m}{1000}$

$m = 1120g$

↑ solution

$\text{mass}_{solu} = \text{mass}_{solv} + \text{mass}_{solute}$

$1120 = X + (1.34 \times 129.6)$

$X = 946.336g$

$\text{molality} = \frac{1.34}{946.336 \times 10^{-3}} = 1.415 \approx 1.42$ m

18. A 1.26 M $Cu(NO_3)_2$ solution has a density of 1.19 g/cm³. What is the weight percent $Cu(NO_3)_2$ of the solution?

%

A. 1.88% B. 2.36% C. 10.5% D. 14.3% E. 19.9%

$V_{solution} = 1L$

$n_{solute} = 1.26$ mole

mass solute $= 1.26 \times 187.56$

$= 236.3256g$

$m = 1000 \times 1.19 = 1190g \leftarrow$ solution

$\% = \frac{236.3256}{1190} \times 100\%$

$= 19.859\%$

$= 19.9\%$

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$\frac{n_{solute}}{V(L)}$

19. What is the molarity of a 25.0% HCl solution if the density is 1.08 g/cm^3?

A. 2.96 M B. 2.70 M C. 7.41 M D. 9.11 M E. 5.49 M

100

25 HCl 75 H_2O

$d = \frac{m}{V}$

$V = \frac{m}{d}$

$= \frac{100}{1.08} = 92.59$ mL

$= 0.093$ L

$n_{HCl} = \frac{25}{36.458}$

$= 0.6857$ mol

$M = \frac{0.6857}{0.093}$

$= 7.40556$

≈ 7.41

20. What is the freezing point of a solution containing 4.134 g naphthalene (Molar Mass = 128.2 g/mol) dissolved in 30.0 grams organic solvent? The freezing point of the pure solvent is 53.0 °C and the freezing point depression K_f is 7.10 °C/m.

A. 45.4 °C B. 48.7 °C C. 52.0 °C D. 17.6 °C E. 7.63 °C

$\Delta T_F = K_f \cdot m$

$= 7.1 \times \frac{(4.134 \div 128.2)}{(30 \times 10^{-3})}$

$= 7.63\ C^{o}$

$\Delta T_F = T_{F_{solv}} - T_{F_{soln}}$

$7.63 = 53 - X$

$X = 45.4\ C^{o}$

21. Calculate the vapor pressure in mmHg of an aqueous solution containing 60.0 g of urea (a nonvolatile solute, molar mass = 60.0 g/mol) in 180.0 g of water 75 °C. The vapor pressure of pure water at 75 °C is 290.0 mmHg,

A. 29.0 B. 100. $\boxed{\text{C. 260}}$ D. 190 E. 150

$P_{soln} = X_{solv} P^{\circ}_{solv}$

$= \frac{1}{11} \times 290$

$= 263.63$ mmHg

$n_{urea} = \frac{60}{60} = 1$

$n_{H_2O} = \frac{180}{18} = 10$

22. Which of the following liquids would make a good solvent for iodine, I_2?
A. HCl B. H_2O C. CH_3OH D. CS_2 E. NH_3

non polar

Like dissolve Like

23. Which one of the following compounds would be expected to have the highest solubility in water?

A. $\cancel{CH_4}$ B. CH_3OH C. $\cancel{O_2}$ D. $CH_3(CH_2)_5CH_2OH$

R-OH

24. A solution process will be endothermic if the magnitude of the intermolecular interaction is in the order _____.

A. solute–solvent > solvent–solvent

B. solute–solvent < solvent–solvent and solute–solute

C. solute–solvent > solvent–solvent and solute–solute

D. solvent–solvent < solute–solute

25. If the pressure of a gas over a liquid increases, the amount of gas dissolved in the liquid will

A. increase

B. decrease

C. remain the same

D. depend on the polarity of the gas

26. The solubility of nitrogen gas at 25°C and a nitrogen pressure of 522 mmHg is 4.7×10^{-4} mol/L. What is the value of the Henry's Law constant in mol/L·atm?

A. 6.8×10^{-4} mol/L·atm
B. 4.7×10^{-4} mol/L·atm
C. 3.2×10^{-4} mol/L·atm
D. 9.0×10^{-7} mol/L·atm
E. 1.5×10^{3} mol/L·atm

$S_g = K P_g$

$4.7 \times 10^{-4} = K \times 0.6866$

$K = 6.84 \times 10^{-4}$ mol/L·atm

1 atm = 760 mmHg

X = 522 mmHg

$X = 0.6868$ atm

Chapter 13 ✓